

Between states and events

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1 Introduction¹

Departing from generally known insights into the distinction between events and states (see also Ramchand, this volume), this chapter addresses borderline cases of predicates that show both stative and eventive behavior and have been difficult to fit within known classifications, such as Vendler’s (1957) distinction between states, activities, accomplishments, achievements, and similar classifications. We propose an analysis of these hybrid predicates, exploring the idea that stativity and eventivity are multidimensional concepts.

The chapter is structured as follows. §2 revisits diagnostics that have been proposed to distinguish states from events, introduces the concept of aspectually hybrid predicates, and characterizes the dimensions we take responsible for a predicate to be interpreted as a state or as an event. §3 provides an introduction to Russian aspect morphology and shows how different perfectivizing prefixes are sensitive to the (un)availability of certain eventive dimensions with stative predicates. §4 discusses various ways in which lexical statives appear to be interpreted as events at the sentence and discourse level. §5 describes stative uses of predicates that are otherwise classified as eventive.

2 States or events? Aspectually hybrid predicates

Vendler (1957) discusses several empirical facts for English that in subsequent literature have been used as diagnostics for states, also cross-linguistically. States are incompatible with the progressive, as they do not involve a process going on in time (1-a). Further tests show that states lack control/agentivity: They are incompatible with agent-oriented adverbs (1-b) and other constructions diagnosing intentionality and/or agentivity (Dowty, 1979; Rothstein, 2004). For instance, they cannot occur as complements of *persuade* (1-c), in imperatives (except *wish*-type uses, see Condoravdi & Lauer 2012) (1-d), and in pseudo-cleft *do*-constructions (1-e). Finally, states cannot be embedded in perception reports (1-f) (see Maienborn, 2005), again unlike events (see Higginbotham, 1983; Parsons, 1990).

- (1) a. #I am knowing/loving Czech.
- b. #I know/love Czech deliberately/carefully.
- c. #Mila persuaded Shakuntala to know/love Czech.
- d. #Know/love Czech!
- e. #What Mila did was know/love Czech.
- f. #I saw her knowing/loving Czech.

Other tests diagnose stativity via the readings obtained in specific contexts. The simple present with states does not (have to) have a habitual reading, unlike what we find with events. Another test concerns the reading of *must*: It is only when the embedded predicate is stative that *must* can have an epistemic reading as opposed to root readings (Condoravdi, 2002) (see also discussion in Arregui, this volume). For instance, (2-a) can (and in fact preferably does) mean that it is highly likely that Mila knows or loves Czech, while (2-b), where *must* embeds an event, can only mean

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that Mila has the obligation to eat garlic soup. In order for events to receive an epistemic reading under *must* they have to be used in the progressive (3). The output of the progressive, in turn, at least under some analyses, has been analyzed as a derived state; we will come back to this notion in §4.

- (2) a. Mila must know/love Czech. (must + state)
b. Mila must eat garlic soup. (must + event)
- (3) Mila must be eating (garlic soup).

A first issue that arises with these tests is that they are almost all “non-stative tests”: They diagnose stativity via infelicity, and therefore do not always allow one to distinguish states from other categories failing these tests too, but possibly for independent reasons. A second and related issue is that not all events pass these tests either: For example, achievements are incompatible with (or dispreferred in) the progressive (cf. Vendler, 1957). A third issue, and the one we concentrate on in this chapter, is that not all statives behave alike with respect to these diagnostics.²

Rather than seeing these issues as shortcomings, we argue that when viewed more closely, stativity tests reveal a fantastic heterogeneity among statives. A similar variation has been observed for the results of agentivity tests: Martin (2022) proposes that the concept of agentivity is a multidimensional concept (Sassoon 2013), and that different agentivity tests pick up different agentive dimensions (such as effectivity, control, foreknowledge, volition). On this view, agentivity is scalar: The more agentive dimensions an event satisfies, the closer it is to the prototype of an action.

In this chapter, we propose to extend this view to the concept of eventivity, as events have been characterized by different dimensions (Maienborn 2005; Copley & Harley 2015; Copley 2019; a.o): transience, extendedness, change, and input of energy (forces). We argue that the more eventive dimensions an eventuality satisfies, the closer it is to the prototype of an event. The heterogeneity in “hybrid” or “atypical” statives can then similarly be linked to the fact that statives differ from each other in the eventive/agentive dimension(s) they satisfy. To these four eventive dimensions, we add (agentive) control, a key dimension of agentive events, because “prototypical event representations contain an agent” (Gärdenfors 2014, 161), and also because agentive control has been used to characterize atypical, “controlled” states (Krifka 2012). These five dimensions yield 32 possible combinations, of which most are likely ruled out because of implications between these dimensions (see discussion in the following subsections). Table 1 gives examples of predicates sorted along the eventive dimensions they satisfy.

predicate	change?	energy?	transient?	extended?	controlled?
<i>intelligent</i>	✗	✗	✗	✗	✗
<i>hungry</i>	✗	✗	✓	✗	✗
<i>live</i>	✗	✗	✓	✓	✗
<i>lose</i>	✓	✗	✓	✗	✗
<i>sleep</i>	✗	✓	✓	✓	✗
<i>fall</i>	✓	✗	✓	✓	✗
<i>float</i>	✓	✓	✓	✓	✗
<i>sit_{ag}</i>	✗	✓	✓	✓	✓
<i>dance</i>	✓	✓	✓	✓	✓

Table 1: Subtypes of English predicates sorted along the eventive dimensions they satisfy

In the following we define the dimensions and discuss some of the constructions that are sensitive to them; see Table 2.³

²This heterogeneity remains also if we make subdivisions of statives into, e.g., stage-level and individual-level predicates (cf. Condoravdi, 1992; Kratzer, 1995), or Davidsonian and Kimian states (in the sense of Maienborn, 2005), two kinds of distinctions we mostly set aside in this chapter. This is so because each subclass might behave uniformly in the application of one test, but not when it comes to all tests.

³‘□’ in Table 2 indicates the eventive dimensions that must be satisfied for a predicate to be compatible with a given construction; in some cases, one out of two dimensions suffices for compatibility with a construction, which we express via parentheses. The Russian data will be discussed in §3.

The English *do*-test also requires effectivity, a core dimension of agentivity.

construction	change	energy	transient	extended	controlled
English progressive			☐	☐	
English <i>do</i> -test		☐	☐	☐	
German <i>geschehen</i>	☐		☐		
French progressive	(☐)		☐	☐	(☐)
Russian delimitative <i>po</i> -			☐	☐	
Russian ingressive <i>za</i> -	☐		☐	☐	

Table 2: Diagnostics for eventive dimensions across languages

2.1 Transience

Events are typically transient (episodic), while “prototypical states” are permanent. However, many statives are transient like events. Transience is often taken as one of the dimensions distinguishing, among statives, stage-level predicates (SLPs) (*be hungry*) and individual-level predicates (ILPs) (*love Czech/be intelligent*; cf. Kratzer 1995). While ILPs come with a default inference of temporal persistence, SLPs do not (Condoravdi 1992; see also Pryslopska 2023, ch. 9 for experimental support). SLPs behave differently from ILPs and pattern with event predicates in many respects in English (Kratzer 1995) and across languages.

2.2 Extendedness

Events are typically extended and last more than one instant, whereas “prototypical states” are true at any instant. This makes achievements (event boundaries under Piñón’s 1997 view) atypical events, and their status as events an object of debate (cf. Piñón, 1997). Above, we already noted that they share affinities with states in the application of some of the tests.

Extendedness is the dimension characterizing a class of statives that Dowty (1979) calls interval states. These include posture verbs like *sit*, *stand*, *lie*, perhaps the most well-known example of “atypical” statives, as they, for example, are compatible with the progressive and do not give rise to an ongoing reading with the simple present (4).

- (4) a. Miranda is sitting/standing/lying on the couch (right now).
b. Miranda sits/stands/lies on the couch (#right now).

Dowty argues that statements like those in (4-a) are only true of intervals: The entity must remain stationary for at least a short period (while *Miranda is on the couch* may be true even if Miranda is continuously in motion).⁴

While posture verbs under their literal meaning usually require animate subjects with the right anatomy for the type of posture, there are also non-literal uses with other subjects (5) (cf. Fraser, 2022). This either indicates that agentive control is not lexically encoded by these verbs (if we assume that both literal and non-literal uses share the same lexical core), or it is, and the non-literal use involves lexical bleaching and the loss of the agent control dimension.

- (5) a. Your book is lying on the table.
b. The water bottle sat *{on the floor/empty}. (from Fraser, 2022, 4)
c. The whale was sitting #(motionless/in the river’s mouth). (from Fraser, 2022, 125)

The fact that the English progressive is compatible with extended (transient) statives (like posture verbs) but not with non-extended SLPs (e.g. *hungry*) suggests that it requires extendedness (and thereby transience).

Extendedness does not suffice to make the counterparts of these English predicates compatible with the progressive in French (see (6)) or Mandarin (see Paul & Yan 2024, 31, ex. 6b built with *zuò* ‘sit’), however.

- (6) #Le verre est en train de reposer sur la table.
the glass is in TRAIN of rest.INF on the table
Intended: ‘The glass is resting on the table.’ (French)

Thus, there is cross-linguistic variation, at least in the application of the tests (in this case the semantics of the progressive), but possibly also in the lexical inventory of verbs (e.g. Talmy 2000 assumes that there are no true posture verbs

⁴See also the division into static and dynamic states in Taylor (1977); Bach (1986).

in French, a verb-framed language).

2.3 Dynamicity: change, energy

Dynamicity is often taken as a distinctive property of events. Dynamicity is either defined via the notion of change or via the notion of input of energy (i.e. forces). Change and force are different notions. A predicate of force does not need to entail change, since one can exert a force against an object that does not move; see discussion in Copley (2019, 107-109) and references therein. Whether it is in principle possible to have predicates of change that do not entail a force is less discussed. The answer will depend on the definition of predicates of force one adopts. If they are defined as predicates expressing an input of energy generated or exerted by the subject's referent, predicates of change do not necessarily convey forces.⁵ For instance, achievements convey a punctual transition between two states-of-affair and thus change, but the subject's referent does not exert energy in this transition, under the reasonable assumption that any energy generation takes more than an instant. Another relevant case are unaccusative predicates like *fall* or *decay*, which express a change, but their subject does not generate energy and is completely passive.

Copley & Harley (2015) and Copley (2019) argue that verbs of posture but also verbs of maintaining, e.g. English *stay*, *remain*, *keep*, describe forces and are therefore eventive. Similarly, Warglien et al. (2012) argue that while pure statives denote degenerate (identity) vectors, predicates like *sit/stand/lie* have a “balancing” force vector. Others maintain that all these verbs are stative, because they lack change (e.g. Levin & Rappaport Hovav, 1995). We observe that verbs of remaining, but also sleeping, behave like posture verbs (and could thus be subsumed under the notion of interval states), in that they involve the dimensions extendedness, transience, and energy input (with further differences with respect to agentive dimensions).

The relation between the notions of force and extendedness has not been very much discussed. It seems that the former entails the latter, since energy generation needs at least a short period of time to take place. But the opposite does not necessarily hold, i.e. interval state predicates are not automatically force predicates. English *live*, for instance, describes interval states in the progressive, but does not convey an input of energy. This likely explains that contrary to posture verbs, it gives rise to an ongoing reading with the simple present, see (7-a). Posture verbs and *live* also differ with regard to the *do*-test, see (7-b).

- (7) a. Hamida lives in Nantes right now.
b. What Rachida did was {*lie/remain*} on the sofa / *#live* in Brussels}.

While extendedness and transience (with or without input of energy) suffice for verbs to pass the progressive test in English, these eventive dimensions still do not suffice for progressivization in other languages such as French (see (8-b) below); here, still more eventive or agentive dimensions are required for the progressive to be acceptable. For instance, *rester* ‘remain’ only appears in the progressive if the subject is agentive (see the contrast in (8), from Martin 2008). This allows such verbs to describe what is sometimes called “agentive states” (“controlled states” in Krifka 2012).

- (8) a. Je suis en train de rester sur un PC à lire des posts sur des forums de fous.
I am in TRAIN of stay.INF on a PC to read.INF INDEF.PL posts on INDEF.PL forums of crazy.PL
‘I am staying on the computer reading silly posts on crazy forums.’
b. #L'autocollant ne veut pas partir, il est en train de rester sur le PC.
the.sticker not wants NEG leave it is in TRAIN of stay.INF on the PC
‘The sticker doesn't want to go, it is staying on the computer.’ (French)

English and German ‘float’ (and similar) is another verb of maintaining. According to Geuder & Weisgerber (2006, 136), it describes an “equilibrium of buoyancy and gravitational forces”, but in contrast to ‘remain’/‘stay’, “movement [thus change] can be freely added”. This makes ‘float’ even more eventive than other verbs of maintaining. When (potential) change is added to extendedness and transience, the progressive is licensed with French *flotter* ‘float’, including with non-agents (9).

- (9) Le verre est en train de flotter sur l'eau.
the glass is in TRAIN to float.INF on the.water
‘The glass is floating on the water.’ (French)

⁵See Wolff et al. (2009) on the relevance of energy generation for the acceptability of inanimate and non-agentive DPs in subject position across languages.

In other cases, energy input can be absent, but change has to be added to extendedness and transience in order for the French progressive to be acceptable, e.g. with *tomber* ‘fall’ (10). Finally, since prototypical activities (e.g. *danser* ‘dance’) generally involve change and control, they are always acceptable in the French progressive.

- (10) Muriel est en train de tomber.
Muriel is in TRAIN to fall
‘Muriel is falling.’ (French)

Thus, the French progressive requires transience, extendedness, and *either* change (*float, fall*) *or* control (*stay*).

2.4 Hybrid events

If the stative vs. eventive distinction is not binary anymore, we expect to find “hybrid events” the same way we found “hybrid states”. Hybrid events are event predicates lacking some eventive dimension(s). Take, for instance, *wait* and *sleep*, which have been treated as events. These verbs often behave like unergatives across languages with regard to tests of eventivity or agentivity (for instance, they pass the *do*-test, see Cruse 1973, suggesting that they convey a force and that their argument is an effector). On the other hand, Maienborn (2005) classifies them as (Davidsonian) states, because they cannot serve as antecedents of expressions like *geschehen* ‘to happen’ in German; see (11) (after Maienborn 2005, 285).

- (11) a. Die Kerze flackerte. Das geschah während...
the candle flickered that happened while
‘The candle flickered. That happened while...’
b. Heidi schlief. #Das geschah während...
Heidi slept that happened while
‘Heidi slept. That happened while...’ (German)

Presumably, ‘happen’ requires change, while events denoted by verbs such as ‘wait’ or ‘sleep’ do not involve change, resembling states on this respect. As for achievements, these are compatible with German ‘happen’, precisely because they involve change (in addition to transience).

- (12) Tamara verlor ihren Schlüssel. Das geschah auf ihrem Weg zur Arbeit.
Tamara lost her key that happened on her way to work
‘Tamara lost her key. That happened on her way to work.’ (German)

On the other hand, since achievements are event boundaries, they do not involve extendedness, control, or energy, which makes them hybrid events as well.

2.5 Interim summary

In sum, hybrid states differ from each other by the eventive and/or agentive dimensions they can instantiate, and “non-stative” tests found across languages are sensitive to different (combinations of these) dimensions. Obviously, the more eventive/agentive dimensions a stative predicate instantiates, the more distant it is from “typical stateness” (temporal persistence, absence of energy or change, passivity, etc.). On such a view, the stative vs. eventive distinction is not binary anymore, but rather scalar in essence, an idea also explored in Fábregas & Marín (2017, §2), Copley (2019), Fábregas & Rodríguez (2020, §5). The nature of the relevant agentive and eventive dimensions as well as the question of whether there is a hierarchy in the dimensions required by non-stative tests within and across languages (i.e., whether a non-stative test requires agentive/eventive dimension D_n necessarily also requires some other dimension D_m , etc.) remains to be investigated in detail.

The overview above already pointed to some potential implications between dimensions: 1) input of energy implies extendedness and 2) extendedness implies transience (see also Dowty 1979, 176–179). Other implications are plausible: 3) agentive control implies extendedness (since exerting agentive control requires a minimum time interval) and thereby transience. Furthermore, controlled (agentive) statives are dynamic, meaning that 4) agentive control implies either change or input of energy. Finally, 5) change implies transience. Taking into account these five implications rules out all but 9 of the 32 possible combinations of the five dimensions, each of which is illustrated by a predicate in Table 1.

The question whether hybrid predicates are states or events, then, might be less relevant, if these are scalar notions to begin with. If one nevertheless wants to answer this question, it very much depends on whether some dimension or another is defined as a necessary and sufficient dimension for events or states. For Maienborn (2005), for example, change seems to be a core dimension of events (if we are correct that non-dynamicity is the source of the incompatibility with ‘happen’). For Copley & Harley (2015), in turn, energy input plays the same role. We assume that events are necessarily dynamic, so they have to involve change or energy (or both). On the flip side then, a state is a predicate that lacks both change and energy. Returning to Table 1, this leads to a division into states (*intelligent, hungry, live*) and events (*lose, sleep, fall, float, sit, dance*).

In the following section, we turn to Russian aspect morphology. We show how different dimensions lead to different perfective meanings with hybrid predicates, and that this has morphological reflexes in this language.

3 Russian aspect morphology and eventive dimensions

Russian, like other Slavic languages, is a language with a grammatical category aspect, i.e. imperfective (IPFV) vs. perfective (PFV), which appears in both the past and non-past domain and even with non-finite verb forms, such as the infinitive and (some) participles. We adopt a standard aspect semantics: With the PFV, the event time is included in the reference time (e.g. Klein, 1995; Grønn, 2004; Altshuler, 2012; Gehrke, 2023), and with the IPFV, the reference time is included in the event time (e.g. Altshuler, 2012; Gehrke, 2023). It is generally taken for granted that the (I)PFV distinction requires some notion of unfolding in time. Under the simplistic view we argued against in §2 that there is a uniform class of states, this would lead to the expectation that statives should not participate in the (I)PFV distinction. However, this is not the case, and we take this as a further indication that stativity is a scalar, rather than absolute concept: Almost all verbal predicates can appear in the PFV aspect or can be perfectivized.

Traditionally, it is assumed that many Russian predicates come in aspectual pairs, which do not differ in their lexical semantics but only in grammatical aspect. In many aspectual pairs, apart from those formed by suppletion, there is a morphological link between the two partners, with either the IPFV being morphologically derived from the PFV by means of a suffix (13-a)–(13-b), or the PFV being morphologically derived from the IPFV, mostly by means of a prefix (13-c)–(13-d).

(13) ASPECTUAL PAIRS IN RUSSIAN

- | | | |
|----|--|---------------------|
| a. | PFV <i>otkryt'</i> > IPFV <i>otkryvat'</i> | ‘to open, discover’ |
| b. | PFV <i>dat'</i> > IPFV <i>davat'</i> | ‘to give’ |
| c. | IPFV <i>čitat'</i> > PFV <i>pročitat'</i> | ‘to read’ |
| d. | IPFV <i>videt'</i> > PFV <i>uvidet'</i> | ‘to see’ |

Accomplishment and achievement predicates regularly form aspectual pairs (the PFV commonly signals that a telos was/is/will be attained); there is a tendency for such predicates to be “born” as PFVs, which are often (e.g. (13-a)) but not always (e.g. (13-b)) lexically prefixed, and for their IPFV counterparts to be derived by means of a suffix. In contrast, there is a tendency for states and activities to be “born” as simple IPFVs, which can be perfectivized by means of a prefix. Some such prefixes are traditionally assumed to derive an aspectual partner (e.g. (13-d)). In other cases, IPFV states/activities behave like *imperfectiva tantum*, which can nevertheless be perfectivized in particular contexts (deriving *perfectiva tantum*), by means of so-called Aktionsart prefixes that provide different kinds of perfective meanings, independently of the lexical base they apply to (cf. Isačenko, 1962). These prefixes have in common that they add temporal bounds to otherwise unbounded eventualities. In this chapter, we focus on ingressive *za-* and delimitative *po-*.⁶

For example, with verbs that have been shown for English to sometimes be interpreted as activities, sometimes as accomplishments, such as incremental theme verbs (and also many degree achievements), we see this variable behavior clearly in the morphology: The basic verb that is used to describe, for instance, a writing eventuality, is a simple IPFV (similar for ‘read’; see (13-c)). Only transitive ‘write’ (in combination with an object that is understood as quantized) is assumed to enter an aspectual pair via a particular prefix (14-a);⁷ in its intransitive use, ‘write’ behaves like an

⁶Other such (temporal) perfectivizing prefixes express the perdurative, evolutive, egressive Aktionsart (among others). In the theoretical literature on Russian (and other Slavic languages) it is common to differentiate between internal/lexical prefixes, which derive new lexical items (e.g. *ot-* in (13-a)), and external/superlexical ones like these Aktionsart prefixes (see discussion in Gehrke, 2008, and references cited therein).

⁷The prefix in this case is usually treated as an “empty” prefix, because it expresses a meaning that is already inherent in the unprefixed IPFV, in this case *na-* ‘on’ (one usually writes something on some surface). With other incremental verbs the prefixes are, for instance, *pro-* ‘through’ (13-c),

imperfectivum tantum, i.e. like an activity predicate (14-b), which can, however, be perfectivized by, e.g., delimitative *po-* (14-c), which adds temporal bounds on both sides of the eventuality.

- (14) a. IPFV *pisat'* *pis'mo* > PFV *napisat'* *pis'mo* 'to write a/the letter'
 b. IPFV. TANTUM *pisat'* 'to write'
 c. DELIMITATIVE PFV. TANTUM *popisat'* 'to write (for a while)'

To see how Russian fits the picture we outlined in §2, we now take a closer look at lexical meanings that are commonly assumed to be expressed by states cross-linguistically.

3.1 Posture verbs

In §2 we saw that posture verbs are extended (and thereby transient) states. In Russian, the posture verbs *sidet'* 'sit', *stojat'* 'stand', *ležat'* 'lie' are traditionally assumed to be *imperfectiva tantum*. They can combine with delimitative *po-* to become temporally bounded (15-a). In this respect they pattern with activities that are *imperfectiva tantum* (15-b).⁸

- (15) a. Ona {*po-stojala* / *po-sidela* / *po-ležala*} tam (neskol'ko vremeni).
 she PO-stood.PFV PO-sat.PFV PO-lay.PFV there some time
 'She stood/sat/lay there (for some time).'
 b. Ona {*po-pela* / *po-plakala* / *po-kričala*} (neskol'ko vremeni).
 she PO-sang.PFV PO-cried.PFV PO-shouted.PFV some time
 'She sang/cried/shouted (for some time).'

We assume that the relevant dimensions for delimitative *po-* to be applicable are therefore extendedness (and thereby transience). However, since *po-* also needs a homogeneous input (see, e.g., Filip, 2003), the predicates that this prefix applies to cannot involve a change of state.

On the other hand, IPFV activities can also combine with the perfectivizing ingressive prefix *za-*, which marks the initial boundary of a process (16-a), but this prefix (with this meaning) is ungrammatical with 'sit'/'stand'/'lie' (16-b).

- (16) a. Ona vošla v komnatu i {*za-pela* / *za-plakala* / *za-kričala*}.
 she in.walked in room.ACC and ZA-sang.PFV ZA-cried.PFV ZA-shout.PFV
 'She entered the room and started to sing/cry/shout.'
 b. *Ona vošla v komnatu i {*za-stojala* / *za-sidela* / *za-ležala*} tam.
 she in.walked in room.ACC and ZA-stood.PFV ZA-sat.PFV ZA-lay.PFV there
 Intended: 'She entered the room and started to stand/sit/lie there.'

Thus, posture verbs do not fully behave like activities either. We assume that the relevant dimensions for ingressive *za-* to be applicable are extendedness (and thereby transience), and change in addition.

3.2 Further potentially hybrid states or events

In §2.4, we discussed 'wait' and 'sleep' as instances of hybrid events, which instantiate the eventive dimensions extendedness, transience and energy, but lack change (and control in the case of 'sleep'). Russian *ždat'* 'wait' and *spat'* 'sleep' are *imperfectiva tantum* that behave like posture verbs: They are compatible with delimitative *po-* (signalling extendedness and thus transience), but incompatible with ingressive *za-* (signalling lack of change). In particular, there is no *zaždat'*, and *zas(y)pat'* is a different lexical item ('fall asleep') in an aspectual pair. Another verb that behaves like 'wait', 'sleep' and posture verbs in this respect is *žit'* 'live', which involves the eventive dimensions extendedness and transience but lacks the other dimensions (recall Table 1): There seems to be a *za-*-prefixed counterpart, but it is rather colloquial and expresses a somewhat different lexical meaning 'to begin to live a quiet life'.

In contrast, *padat'* 'fall', which we assume to imply extendedness, transience and change, but to lack energy and control, is compatible with *za-*; this shows that ingressivity does not require energy. Finally, *plavat'* 'float' behaves like other activities in being compatible with both *po-* and *za-*, thus supporting the characterization of such verbs in Geuder & Weisgerber (2006) as optionally implying change, as outlined in §2.3.

vy- 'out' (as in *vy-pit'* 'drink'). There is debate whether "empty" prefixes should be treated as internal prefixes (e.g. Gehrke, 2008), or not; since Isačenko (1962) only views suffixation as deriving "true" aspectual pairs, he subsumes all "empty" prefixes under the resultative Aktionsart.

⁸Examples like those in (15) also show that perfectivity cannot be the same as telicity, since these PFVs are compatible with durative 'for'-adverbials (and not with 'in'-adverbials), thus passing a typical test for atelicity.

Other stative verbs seem to be incompatible with either delimitative *po-* or ingressive *za-*, e.g. *stoiť* ‘cost’ and *izlučat* ‘radiate’. This indicates that these verbs behave like “prototypical states” in lacking any eventive dimension. Intuitively, this makes sense for ‘cost’, but is unexpected for ‘radiate’. However, it is possible that there is a morphological rather than semantic explanation for the lack of delimitative and ingressive forms with this verb: There is already a prefix (*iz-*) deriving a verb from *luč* ‘light’.

In §2.3 we also saw that there is disagreement in the classification of verbs of maintaining as either stative or eventive. Russian *osta(va)t’sja* ‘stay, remain’ forms a regular aspectual pair, in which the PFV is the basic form. The morphology thus suggests that it is lexically eventive.

True copular constructions (e.g. ‘be intelligent/hungry’) are always IPFV, because *byt’* ‘be’ is an *imperfectivum tantum*.⁹ However, there is a difference between long and short form adjectives in predicative position, which sometimes (but not always, cf. Geist 2010) signals whether the state is transient (short form) or not (long form):

- (17) a. Ekaterina zdorova.
Ekaterina healthy.SF.F.SG
‘Ekaterina is healthy (right now).’
b. Ekaterina zdorovaja.
Ekaterina healthy.LF.F.SG
‘Ekaterina is healthy (permanently).’

Finally, Russian can also derive a verb from the adjective ‘hungry’, *golodat* ‘starve, be hungry’, which is a transient state. This verb is incompatible with ingressive *za-* (indicating a lack of change), but compatible with delimitative *po-* (indicating extendedness and thereby transience).

3.3 Interim summary

Our brief discussion of Russian verbal morphology has shown that simple (i.e. non-prefixed) IPFVs describe a non-bounded event type (a state or activity) or an event type with variable telicity (e.g. with incremental verbs). Prefixes can provide particular bounds for these, which can be temporal initial bounds (ingressive *za-*), final bounds (e.g. egressive *ot-*; cf. Isačenko 1962), or both (e.g. delimitative *po-*); with incremental verbs on their telic reading, particular prefixes signal resultativity. Different stative predicates behave differently with respect to the perfectivizing prefixes they are compatible with, which in turn gives rise to different perfective meanings they acquire in these cases (e.g. ingressivity, temporal boundedness). The claim we made in this section is that restrictions on the types of PFV readings we get relate to different dimensions of eventivity/stativity. PFV states and activities, then, are derived events, which is also supported by the fact that such verb forms lead to narrative progression in discourse (cf. Gehrke, 2022). Nevertheless, we argue that as basic predicates they are still “just” states and activities (even if in some cases hybrid states or activities), and in the following section, we will take a closer look at PFV states.

4 Deriving eventivity with states

The notions “event” and “state” have also been employed as a dichotomy in works that investigate the behavior of verbal predicates, aspects and tenses in discourse, to define discourse or rhetorical relations between stative and eventive discourse referents (e.g. Kamp & Reyle 1993; Lascarides & Asher 1993; ter Meulen 1995; see also de Swart this volume for further discussion). For example, Kamp & Reyle (1993) argue that events and states introduce an event and state variable, respectively, with events being included in a time and following a previously introduced event (leading to reference time movement), and states always overlapping with a time (and thus also with previously introduced events). This accounts for the long-known fact that eventive sentences “move the narration forward”, whereas stative sentences do not.¹⁰ In these works, events and states are not taken as lexical primitives but as derived entities at the discourse level, without specifying how the relevant interpretation as a state or an event comes about compositionally.

A more compositional approach is found in de Swart (1998) who defines three basic eventuality types, states, processes (i.e. activities), and events (i.e. accomplishments and achievements), based on the properties homogeneous vs. quantized ($\sim$ atelic vs. telic) and stative vs. dynamic. These eventuality descriptions are then proposed to be the input to aspectual operators, including coercion operators, followed by Tense. For example, the progressive operator is

⁹There are also semi-copulae in aspectual pairs (e.g. *okaz(yv)at’sja* ‘seem’, *sta(va)t’* ‘become’), with which the state is transient.

¹⁰But see Martin (2008, ch. 5) and Altshuler (2021, §5) for discussion of apparent exceptions.

argued to take a process or event as an input and to return what we call a derived state. Treating progressives as derived states is a common practice in the literature, and accounts for the behavior of progressives in discourse (see also Asher, 1993; de Swart, this volume).

Let us then see under which circumstances statives can be interpreted as derived events, or come with an additional event inference. Many European languages make an aspectual distinction in the past tense domain between imperfect and aorist (under various labels). In discussing French and Bulgarian data, de Swart (1998) treats these as aspectually sensitive tenses. She argues that the French *imparfait* is a past tense operator that requires a homogeneous input, while the *passé simple* is a past tense operator that requires a quantized input. If the input is not of the right type, coercion operators are employed, and in the latter case we obtain a derived event. For example, the *passé simple* applies to a stative predicate like ‘know’, a h(omogeneous) predicate, only after a coercion operator turned it into a (derived) e(vent) (C_{he}), as in (18) (from de Swart, 1998, 370).

- (18) a. (Soudain,) Jeanne sut la réponse.
suddenly Jeanne know.PS the answer
‘(Suddenly,) Jeanne knew the answer.’ (French)
b. [PAST [C_{he} [Jeanne know the answer]]]

The precise interpretation that the coercion operator gives rise to is determined by context: In (18), it is usually assumed that the adverb ‘suddenly’ triggers an inceptive reinterpretation. In such contexts Russian would not employ ‘know’ in the PFV (as an inceptive state) but rather switch to a different lexical item with the meaning ‘find out’ (a lexical achievement) (19).

- (19) Vnezapno Žanna uznala otvet.
suddenly Žanna found.out.PFV answer
‘Suddenly, Žanna found out the answer.’ (Russian)

We can also add to de Swart’s observations about French that the adverbial ‘suddenly’ is required for the inceptive reading to arise – without any adverbial, (18-a) triggers the inference that Jeanne *said* the answer, which is an actualistic interpretation. We come back to this point shortly.

In contemporary French, the *passé simple* is confined to formal, written texts, and its perfective function has been taken over by the French perfect (i.e. the *passé composé*) otherwise. Therefore, in the following we mainly employ the *passé composé* to illustrate the effect that perfectivity can have with French stative predicates. A first case in point is the actuality inference reported for modal verbs in their root readings when combined with some tense/aspect morphology (Bhatt 1999, Hacquard 2010; see also discussion in Arregui this volume; Owusu this volume). For instance in French, the *passé composé* sentence (20), despite being built with a (stative) modal verb, entails or at least strongly implicates that some event satisfying the property described in the prejacent happened. When the *imparfait* is used instead of the *passé composé*, no event inference is triggered.

- (20) Soumia {pouvait / #a pu} soulever le frigo, mais elle ne l’a pas fait.
Soumia can.IMPF has could lift the fridge but she NEG it.has NEG done
‘Soumia could lift the fridge, but didn’t do it.’ (French)

The same observation can be made for Russian (21).

- (21) Sumija {mogla / #smogla} podnjat’ xolodil’nik, no ne sdelala ètogo.
Soumia could.IPFV could.PFV lift.INF.PFV fridge but not did.PFV this
‘Soumia could lift the fridge, but didn’t do it.’ (Russian)

It has been argued that the actuality inference in (20) results from a clash between the semantics of the *passé composé* and the stativity of the root modal: The *passé composé* needs a bounded eventuality (similar to what de Swart 1998 argued for the *passé simple*), what the ability cannot plausibly be (Mari & Martin 2007). Homer (2021) proposes an account in terms of coercion: In the absence of any temporal adverbial, coercion must be actualistic. The covert operator, ACT, brings in an event actualizing the modal state, as in (22). On this view, sentence (20) (without the continuation) is an example of an aspectually hybrid statement: It asserts the existence of an event e of Soumia lifting a fridge, such that e is simultaneous with some state of Soumia being able to lift a fridge.

- (22) [PAST [ACT [pouvoir [Soumia lift the fridge]]]] (simplified)

The reinterpretation does not have to be actualistic though. In particular, when we add a temporal adverbial such as *pendant un certain temps* ‘for a certain time’ to (20), the actuality inference is much weaker, and the second clause is felt less contradictory than in (20) (Mari & Martin 2007, Homer 2021). Under Homer’s account, adverbials trigger different kinds of repair to the aspectual mismatch between the *passé composé* and the modal stative predicate, each mustering a specific covert operator different from ACT (e.g. MAX with ‘for a certain time’, INCH [inceptive] with ‘suddenly’), thus avoiding the actuality inference.

Homer (2021, §2.2.3) proposes to account in the same way for the event inference of non-modal stative predicates such as *coûter* ‘cost’ (Hacquard 2006, 19), see (23).

- (23) Le pull a coûté trois euros.
the pullover has costed three euros
‘The pullover costed three euros.’
~~ The pullover was bought. (French)

A potential extension concerns the case of attitude verbs such as *savoir* ‘know’ (recall that the variant of (18-a) without the adverbial triggers the inference that Jeanne said the answer), as well as behavior-related adjectives (e.g. *clever*), which also triggers an event inference in *passé composé* sentences (24) (Martin 2008, ch. 3; see also Smith 1997, 195).

- (24) Khadija a été intelligente.
Khadija has been clever
‘Khadija was clever.’
~~ Khadija did something clever. (French)

An issue for Homer’s account is the case of extended predicates like those in (25), where coercion is not actualistic despite the absence of a temporal adverbial.

- (25) Le peintre Henri Matisse {a vécu / est resté à Nice} / a flotté sur l’eau.
the painter Henri Matisse has lived is stayed in Nice has floated on the water
‘The painter Henri Matisse lived/stayed in Nice/ floated on the water.’ (French)

In this context, we note that ‘live’, ‘stay’ and ‘float’ instantiate some eventive dimensions, whereas ‘cost’ and ‘intelligent’ do not and are prototypical states, and this is also what the Russian morphological facts suggest (recall §3.2). It might be possible that actualistic inferences arise only with prototypical states, whereas hybrid ones access an eventive dimension they are lexically equipped with when they combine with perfectivity, which makes them eventive enough for the aspectual operator so that coercion via ACT is not necessary.

In sum, we showed that verbs which are lexically stative can have derived eventive senses. We provided data from two languages with quite different aspectual systems, French and Russian, to demonstrate how perfectivity applied to statives gives rise to different types of derived events. Such cases of “derived” aspectual hybridity, possibly obtained via coercion (e.g. in French), are very different from cases of aspectual underspecification addressed in the next section: The latter concerns verbs that have *underived* eventive and stative uses.

5 The stative/eventive alternation

In §2 we saw that besides well-behaved states satisfying all key properties of statehood (*love/know Czech*), we also find hybrid states of different shades, depending on which eventive and/or agentive dimensions they satisfy. Another related but different phenomenon is illustrated by verbs that are often taken to be inherently eventive but have stative uses as well at the lexical level (that is, the relevant stative use is not derived via the composition with aspectual operators). For example, Gawron (2009) shows that degree achievements, such as *widen*, *redde*, and what he calls path shape verbs, such as *cross*, *zigzag* are often used to describe events, but not so on their extent reading, illustrated in (26) (see also Talmy, 2000; Matlock, 2004; Deo et al., 2013; Delbecq, 2015, for discussion).

- (26) a. The road widens between San Francisco and San Jose.
b. The road runs/zigzags along the coast.

Under the eventive reading, these predicates pass typical tests for events (27), which they fail under the extent reading (28): compatibility with the progressive, habitual reading with the simple present, intentionality/agentivity diagnostics.

- (27) a. The car was crossing the river.
 b. The car crosses the river{regularly/#today at 10}.
 c. The car crossed the river carefully.
 d. Mila persuaded the car to cross the river.
 e. Cross the river, car!
- (28) a. #The bridge was crossing the river.
 b. Look, the bridge crosses the river! (not habitual)
 c. #The bridge crossed the river carefully.
 d. #Mila persuaded the bridge to cross the river.
 e. #Cross the river, bridge!

Gawron (2009) notes that while very few predicates only have an extent reading (e.g. *span*), most predicates with an extent reading also have an event reading.

Other predicates with dynamic and non-dynamic (state-like) uses are verbs like *cover*, *fill*, or *surround* (cf. *The snow covers the mountain*). Kratzer (2000) proposes that both uses involve a causing eventuality, which can be dynamic or stative. In contrast, Rappaport Hovav (2017) argues that the stative use does not involve an external argument: The subject is the figure contained in the location described by the object. In the following, we discuss two types of approaches to event and extent readings that maintain the same event structural and argument structural properties for both uses.

5.1 Atemporal change

Gawron (2009) proposes an underspecification account of event vs. extent readings of the type in (26). While the predicates involved are analyzed as intrinsically dynamic, in the sense that they involve change along a particular axis, they are underspecified with respect to whether the change happens in the spatial or temporal domain (i.e. for the feature [$\pm$ State]).

The proposal to take dynamicity to involve change in either the spatial or temporal domain calls into question the idea that dynamicity always involves temporality (or in other words, what we have labeled “dynamicity” might be a misnomer). Quite generally, it makes sense to analyze the event structures of change-of-state (or -location) predicates not as inherently temporal, but to only be interpreted temporally in some cases and not in others.¹¹ The domain of paths is a good case in point, as already noted by Fong (1997), based on examples like (29).

- (29) a bridge into Kyiv

In (29), we have a directional PP, but there is no change of location in time, only in space.

We see a similar pattern in languages that distinguish between directional (path) and locative (place) interpretations of certain PPs by case-marking, such as German and Russian. In these languages, the complement of prepositions like ‘in’ and ‘on’ is accusative-marked when we are dealing with a directed motion along a path that ends at the location specified by the PP, but marked in an oblique case when we are dealing with a (basic) place interpretation (30) (see Gehrke, 2008, and references cited therein).

- (30) a. Sie tanzte in {der /die} Stadt.
 she danced in the.F.DAT the.F.ACC city
 ‘She danced {in (locative) / into (directional)} the city.’ (German)
- b. Ona {tancevala v gorode / priexala v gorod}.
 she danced.IPFV in city.M.PREP TO.drove.PFV in city.M.ACC
 ‘She {danced in the city (locative) / came to the city (driving) (directional)}.’ (Russian)

When it comes to examples of the spatial change kind, such as (29), we see that German and Russian employ accusative case, i.e. the “directed motion” pattern (31).

- (31) a. eine Brücke in die Stadt
 a bridge in the.F.ACC city
 ‘a bridge into the city’ (German)

¹¹This fits well with the idea that the VP domain is atemporal per se and involves event kinds, whereas temporality only comes in when the VP is embedded under aspect and tense (see, e.g., Gehrke, 2019; Gehrke & McNally, 2019, and references cited therein).

- b. most v gorod
bridge in city.M.ACC
'a bridge into the city'

(Russian)

Russian motion verbs can also give rise to an extent reading, and in (32) we see several morphological and temporal options.

- (32) a. Doroga {idet / proxodit} po poberež'ju.
path.NOM goes.DET.IPFV THROUGH.goes.IPFV along coast
'The path goes along the coast.'
- b. #Doroga xodit po poberež'ju.
path.NOM goes.INDET.IPFV along coast
- c. Doroga {proxodila / prošla} po poberež'ju.
path.NOM THROUGH.went.IPFV THROUGH.went.PFV along coast
'The path went along the coast.'

The first verb form in (32-a) is an unprefix motion verb of the “determinate” type. With unprefix motion verbs, which are always IPFV, Russian has pairs of determinate-indeterminate verbs, with the former necessarily expressing a single motion in one direction, and the latter non-directed or habitual motion or ability (and all specifying a particular manner of motion, e.g. ‘run’, ‘drive’, ‘fly’) (see Gehrke, 2008, and literature cited therein for discussion). The indeterminate counterpart in (32-b) is unacceptable in this case, showing that the extent reading requires unidirectionality. The other verb forms in (32) are prefixed motion verbs, and the prefix itself specifies the direction (the root names the manner). (32-c) is a past tense example, with the only difference in grammatical aspect (we are dealing with a suppletive aspectual pair): With the IPFV, the sentence is reported to be a scenery description, but with the PFV, we get the impression of a sequence of events (e.g. the path went along the coast and then made a turn inland). We will come back to this below.

In sum, the observations concerning (30)–(32) provide further empirical evidence for the claim that the notion of change does not have to be inherently temporal, but can be seen as more abstract. Only when change (along a path) is interpreted in the temporal domain do we get the eventive dimension and an eventive interpretation of the situation as a whole.

5.2 Fictive motion

Still, there might be another way to approach the extent use of motion predicates that does not give up on the intuition that change intrinsically involves temporal dynamicity. Talmy (2000) calls “fictive motion” the use of motion predicates illustrated in (26-b). Matlock (2004) established through experiments that when people process motion predicates in this fictive use, they tend to mentally simulate motion. That is, they construct a spatial model and simulate motion along a path, in analogy to the way they enact motion along a real (as opposed to fictive) path. A possibility that might be worth exploring is that the cognitive simulation process is the reflection of the fact that under uses as in (26-b), motion verbs turn into modal expressions, describing a modal state of affairs. More concretely, a sentence such as *The road zigzags along the coast* could be analyzed as meaning that in all relevant worlds, a motion event which has the road as its path is a zigzagging along the coast event. Under this view, motion predicates still describe motion events in their stative, extent use (which accounts for the presence of directional morphology, for instance), but the events take place in possible worlds.

In this connection, it is interesting to note that in French, fictive motion statements trigger an actuality entailment with the *passé composé*, as observed with modal predicates (Bhatt 1999, Hacquard 2010, recall §4), see (33).

- (33) Le chemin {zigzaguit / #a zigagué} jusqu'à la grotte, mais personne n'a jamais osé l'emprunter.
the path zigzag.IMP has zigzagged until the cave but nobody NEG.has ever dared it take
'The path zigzagged up to the cave, but nobody ever dared to take it.'

(French)

Similarly, we saw in Russian (32-c) that with the perfective aspect, the interpretation is more eventive, since the sentence does not simply describe the scenery but feels like being part of a sequence of events. In (34), switching to a different verb, we see that just like in French (33), the aspectual contrast correlates with whether or not we get an actuality inference, see (34).

- (34) Doroga {petljala / #propetljala} po poberež’ju, no nikto nikogda ne osmelivalsja po
 path.F.NOM meandered.IPFV meandered.PFV along coast but nobody never not dared.IPFV along
 nej projti.
 her go.INF.PFV
 ‘The path meandered along the coast, but nobody ever dared to walk along it.’ (Russian)

Thus, it seems that at least in perfective contexts, extent readings require some actual motion, except perhaps in the presence of a ‘for’-adverbial. Under the first analysis option outlined in §5.1, we would have to say that in these perfective examples, we are again dealing with coercion (recall discussion in §4).

6 Conclusion

In this chapter, we started out from standard diagnostics proposed in the literature to distinguish stative from eventive predicates. We noted that these tests are sensitive to different dimensions of eventivity and agentivity (transience, extendedness, change, energy input, agent control), which supports a multidimensional analysis of the stativity concept. We proposed that the heterogeneity within the class of statives and in the “non-stative” tests across languages can be traced back to the fact that different “hybrid” statives fulfill different eventive dimensions and different “non-stative” tests require different (combinations of) eventive (or agentive) dimensions.

In subsequent sections, we took a closer look at the effect of perfectivity on stative predicates, primarily in French and Russian, arguing that it gives rise to “derived” events. We also observed that there are certain predicates that are underspecified with respect to the state-event distinction already at the lexical level, such as extent vs. event readings of e.g. motion verbs. We outlined two kinds of approaches to this phenomenon, which both maintain the same argument and event structural properties of these verbs in both readings.

Throughout the chapter, we argued that the fact that stative predicates do not behave uniformly with respect to eventive diagnostics and also give rise to different interpretations in combination with temporal adverbials and perfectivity is best captured by the idea that stativity is a multidimensional concept. We hypothesized that the different dimensions associated with stative predicates ultimately can account for these differences. We hope that the multidimensional approach of the distinction between states and events will be fruitful in future research.

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List of index items (suggestions)

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List of abbreviations

ACC	accusative case
DAT	dative case
DET	determinate
F	feminine
INDEF	indefinite
INDET	indeterminate
INF	infinitive
IPFV	imperfective
LF	long form
M	masculine
NEG	negation
PFV	perfective
PL	plural
PREP	prepositional case
SF	short form
SG	singular